DAEA-Net: Dual Attention and Elevation-Aware Networks for Airborne LiDAR Point Cloud Semantic Segmentation

Yurong Zhu, Zhihui Liu, and Changhong Liu

***Abstract*— Semantic segmentation of airborne laser scanning (ALS) point clouds remains a challenging task due to the complexity and diversity of 3-D scenes in the real world. Currently, most deep learning-based airborne LiDAR point cloud segmentation methods prioritize designing local feature extraction operators while overlooking the long-range dependencies among neighborhoods and the inherently diverse properties of point cloud data. To address these issues, this article introduces a dual-attention and elevation-aware airborne LiDAR point cloud semantic segmentation network (DAEA-Net) built upon an encoding–decoding architecture. First, we develop a cross multiple anti-affine attention (CMAAA) module that effectively captures global contextual information across different neighborhoods through interactive learning of multiple features. Second, we introduce an elevation awareness (EA) module that uses normal vectors to establish a geometric similarity discriminant for each neighboring point. It incorporates an autoencoder architecture to fuse elevation information, enhancing the horizontal structural dissimilarity between objects of similar height while enriching the representation of elevation data. Additionally, to compensate for the potential information loss in the encoding–decoding hierarchical structure, we design a lightweight U-global attention (UGA) module to link decoding and encoding hierarchical levels. It merges features of different resolutions and levels during downsampling and upsampling through pooling while utilizing the self-attention mechanism to enhance the network's global expression capability. The proposed DAEA-Net enhances ALS semantic segmentation performance by enabling interactive learning of multiple features and effectively representing elevation information. Extensive experiments conducted on two datasets demonstrate that our method delivers superior semantic segmentation performance compared to several existing advanced techniques.**

***Index Terms*— Attention mechanism, elevation information, encoder–decoder structure, global perception, point cloud semantic segmentation.**

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I. Introduction

DRIVEN by advances in sensor technology, modern airborne detection systems utilize noncontact methods to directly acquire point cloud data in real space. Compared to traditional satellite images [1], this data can more accurately and intuitively reflect the 3-D features of the real world, providing detailed information on shape, geometric structure, and texture. Propelled by emerging industries such as smart cities [2], powerline detection [3], and 3-D city modeling [4], the need for efficient and accurate airborne LiDAR point cloud semantic segmentation has become crucial for technical support in these fields. However, due to the disordered, unstructured, and unevenly distributed nature of point cloud data, achieving highly accurate semantic segmentation of 3-D airborne LiDAR point clouds remains challenging.

Early point cloud semantic segmentation methods begin by extracting handcrafted features with rich geometric information, such as curvature, normals, edges, and densities. Subsequently, they perform point classification using traditional discriminators like support vector machines [5], random forests [6], and conditional random fields [7]. However, these manual features, which are heavily based on expert knowledge, often overfit to specific datasets, resulting in models with poor generalization capabilities. This issue makes it challenging to meet the real-time demands of point cloud applications. Inspired by significant advances in deep learning, particularly in the image processing domain, researchers have recently developed a variety of deep learning-based semantic segmentation methods specifically for 3-D point clouds. These innovative methods range from voxel-based approaches [8], [9], which transform point clouds into voxel grids, to point-based primitives [10], [11] that work directly with raw points, and graph network-based techniques [12] capturing the topological structure of the data. The prevailing approach now combines the point cloud's inherent geometric features with deep learning models to enhance geometric distinctions between points and significantly improve the model's ability to generalize across different datasets.

Despite the significant success of these methods in point cloud segmentation, several problems or limitations persist.

1) Many existing methods primarily focus on the neighbors constructed by K-nearest neighbor (KNN) or ball queries and on how features are aggregated, but they

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overlook the long-range dependency patterns among different neighbors. In contrast, the Transformer model, a novel neural network architecture, has achieved remarkable success in various image processing tasks, such as those in [13] and [14], demonstrating that long-range correlations are crucial for learning global context.

2) Some current methods incorporate geometric information of point cloud data, such as normals, densities, edges, and elevations, into the network architecture [15], [16], [17]. However, it remains challenging to integrate this information effectively into the network to ensure adequate representation.
3) Most current methods employ the encoding–decoding architecture of PointNet++, which only considers interactions within the same resolution features and lacks the interactive learning between different resolutions. This limitation hinders the model's ability to learn complex structural information. Although some methods [18], [19], [20] use a pyramid-like structure to increase the receptive fields at different network levels and improve model performance, this comes at the cost of a significant increase in the number of parameters.

To address the aforementioned challenges, we propose an airborne LiDAR point cloud semantic segmentation network, DAEA-Net, which is based on dual attention and elevation awareness (EA). DAEA-Net primarily comprises three modules: a cross-multiple anti-affine attention (CMAAA) module, an EA module, and a lightweight U-global attention (UGA) module. The CMAAA module effectively addresses the long-range dependency challenges in local encoding (LE). The EA module enhances the vertical discrimination of objects by leveraging a geometric structure discriminant, local point pair expression, and both relative and absolute elevation representation. It also improves the horizontal discrimination among objects at the same elevation. To fully utilize information across different receptive fields, the lightweight UGA module maps both downsampled and upsampled outputs to the same resolution, using adaptive weighting to enhance the representation of downstream classifier outputs. Extensive experiments on the ALS semantic segmentation task demonstrate the superior performance of the proposed DAEA-Net.

The main contributions of this article are summarized as follows.

1) We developed a CMAAA module to optimize the network's encoding aggregation layer, enhancing the extraction of discriminative and representative global context information. This module processes aggregated updated centroids as objects and interactively fuses the coded features multiple times, effectively compensating for the lack of long-range dependencies in LE.
2) We designed an EA module that utilizes weighted normal vectors and eigenvalues to construct relative representations and geometric similarity discrimination between points. This enhancement enables the model to more accurately determine and describe each point's relative and absolute elevation information, thereby allowing the model to effectively distinguish between objects of varying heights and identify different objects at the same height.
3) We propose a lightweight UGA module that adaptively fuses features from different layers and levels through a self-attention mechanism, effectively resolving the fusion difficulties caused by resolution differences.
4) DAEA-Net achieves lower parameter usage and excellent semantic segmentation performance on two widely used large-scale point cloud datasets.

The rest of the article is organized as follows. Section II provides an overview of the related work. Section III describes the proposed DAEA-Net and its components in detail. Sections IV and V present a comprehensive analysis of the experimental results. Finally, Section VI summarizes our work.

II. Related Work

Point cloud semantic segmentation methods are mainly categorized into traditional machine-learning-based methods and end-to-end deep learning-based methods.

A. Traditional Machine-Learning Methods

Traditional machine learning methods for point cloud segmentation rely on constructing handcrafted features based on the neighborhood of each point, followed by the selection of an appropriate classifier for prediction. These handcrafted features are designed to maintain rigidity, such as invariance to translational and rotational transformations. Typically, this involves transforming the 3-D Euclidean coordinate system into a relative coordinate system or constructing attributes like density. Li et al. [17] further explored this neighborhood relationship by constructing covariance matrices of relative coordinates, where the output eigenvalues determine the significance of the point, line, or plane structures within the point cloud, a technique that has found extensive application in deep learning contexts. Retaining the eigenvector corresponding to the smallest eigenvalue allows for the construction of an important attribute: the normal vector of the point cloud's surface. This facilitates the creation of local feature descriptors for each point, with representative methods including PFH [21], FPFH [22], and SHOT [23]. This robust local characterization has led to a proliferation of methods in tasks such as airborne LiDAR point cloud classification [24], [25], [26].

However, machine-learning-based methods still require significant time and effort to manually extract features for each point. Additionally, manual features struggle to handle point cloud segmentation tasks in real-time, especially in complex scenes.

B. Voxel-Based and Projection-Based Methods

Since point clouds are unordered sets of points, it is impossible to directly apply 2-D convolutional neural network models to point cloud data processing. To address this problem, researchers have converted point clouds into regularized 3-D voxels or projected them into multiple images. They then used 3D-CNNs for feature extraction [8], [9], [27], [28], [29] or

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extracted features using 2D-CNNs [30], [31], [32]. However, voxelizing point clouds may result in the loss of detailed features, and projecting them as a multiview method can disrupt the data's structure. Additionally, the computational effort of both methods increases significantly as the number of point clouds grows.

C. Point-Based Methods

Point-based methods directly use point cloud data, including 3-D spatial coordinates and additional attributes such as intensity and color, as input to the network for feature learning. The primary point-based approaches are convolution-based, graph-based, and attention-based methods.

1) Convolution-Based Methods: The PointNet [10] network provides a global description of the entire point cloud via a max-pooling operation but overlooks local contextual information between points, resulting in poor performance on tasks requiring attention to local features. PointNet++ [33] adopts the concept of a 2-D convolutional architecture, extracting features from local neighbors in the chosen domain via a multilayer perceptron (MLP) with shared weights and aggregating local information through max-pooling. However, its depiction of local neighbors remains coarse, leading to poor robustness in complex scenes. In response, PointCNN [34] introduced a novel transformation operator that normalizes point cloud data through a weighted ordering method, ensuring consistent generation of normalized point features despite changes in input order. This significantly enhances the algorithm's robustness. Thomas et al. [35] extended 2-D convolution into 3-D space by proposing KPConv. KPConv determines the weights of neighboring points based on their Euclidean distance to kernel points and performs convolutional aggregation of point features within the neighborhood. Because the center point's position is not fixed, KPConv is more flexible in processing point cloud data than traditional fixed-grid convolution. To better utilize features of varying resolutions in encoding–decoding architectures, NeiEA-NET [19], MVPNet [20], and others adopt a pyramid-like structure to integrate multiscale features for point cloud segmentation.

2) Graph-Based Methods: To enhance the modeling of edge structures between central and neighboring points, several methods have incorporated graph neural networks to construct a graph for each point, facilitating feature learning. For instance, DGCNN [12] employs an edge convolution operator, EdgeConv, which convolves edge features to better represent local geometric structures. RandLA-Net [36] introduces a local feature aggregation module suitable for large-scale point cloud modeling, preserving geometric details while progressively expanding the perceptual field.

3) Attention-Based Methods: In recent years, Transformers have achieved significant results in various fields. Many researchers have begun to apply the attention mechanism to 3-D point cloud classification and segmentation problems, achieving remarkable performance. PVT [9] extracts coarse-grained local features through a sparse window attention module and aggregates point cloud global features using two self-attention variants to achieve efficient and accurate classification and segmentation. DGACN [37] utilizes the self-attention approach to normalize the constructed geometric point-pair information into weights, modeling finer grained geometric relationships. CSANet [38] designed a cross-attention module that is not only alignment invariant but also interactively extracts coordinate and feature information. 3DGTN [39] proposes pointpatch and channel dual-attention, enhancing the network's ability to learn key information in the spatial and feature domains.

D. Elevation Information-Based Deep Learning Methods

Recent studies have shown that elevation information plays a crucial role in airborne LiDAR point cloud segmentation tasks. GraNet [40] enhances the model's interpretability in the elevation dimension by upscaling elevation information to dimensions corresponding to each layer in the encoding–decoding framework using fully connected layer and achieving feature fusion through summation. GADH-Net [17] employs various elevation preprocessing methods as network inputs, such as Zmax-Zmin, Z-Zmin, and Zmax-Z, to enhance the weight of elevation information in feature learning. DA-Net [41] constructs simple geometric height features by multiplying the negative exponent of relative distances with the z coordinate. Dense-LGEANet [42] maps height information into the weight matrix of an attention mechanism using fully connected layer, enhancing the network's perception of objects. However, these methods' application of elevation information remains relatively simple and fails to fully explore its potential.

III. Methodology

In this section, a detailed description of the DAEA-Net is provided. First, we explain the LE used for representing the point cloud's local structure. Second, the CMAAA and EA are detailed. Finally, the UGA is described to enhance the discriminative capabilities of the downstream classifiers. The overall network architecture is illustrated in Section III-E.

A. Local Encoding

Given a point cloud set $\mathcal{N} = \{p_1, p_2, \ldots, p_N\}$ containing N points, the corresponding prior features are $\mathcal{F} = \{f_1, f_2, \ldots, f_N\}$ in $\mathbb{R}^D$, where the dimension D represents the dimensionality of the features. For each central point p_i, obtained by the farthest point sampling (FPS), we use the ball query algorithm to obtain its K neighbors $\{p_i^1, p_i^2, \ldots, p_i^K\}$.

To enhance the network's learning of the complex geometric structure of point cloud surfaces, previous studies [36], [43], [44] utilized simple edge information, such as relative coordinates and Euclidean distance, as network input. Such edge information remains invariant under the rotation and translation of point cloud data. Consequently, in each encoding layer of the network, relative and Euclidean representations were constructed for spatial coordinates and prior features. Furthermore, as angular information remains largely invariant to rotation and translation, it was also incorporated into geometric LE. The detailed LE structure is illustrated in Fig. 1.

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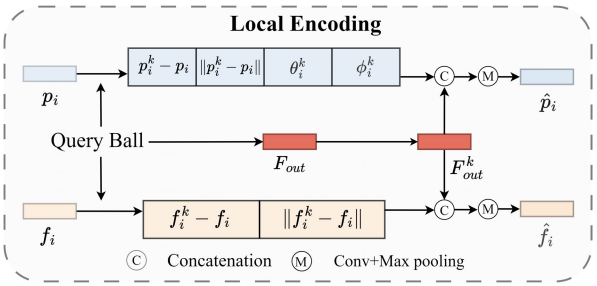


Fig. 1. Architecture of the proposed LE.

The local structure representation of the geometric space is as follows:

$$\hat{p}_i^k = \left(p_i^k - p_i, \|p_i^k - p_i\|, \theta_i^k, \phi_i^k\right) \tag{1}$$

where $p_i^k - p_i = (\Delta x_i^k, \Delta y_i^k, \Delta z_i^k) = (x_i^k - x_i, y_i^k - y_i, z_i^k - z_i)$ represents the relative difference in the coordinate space, and $\theta_i^k = (((\Delta x_i^k)^2 + (\Delta y_i^k)^2)^{1/2})/(((\Delta x_i^k)^2 + (\Delta y_i^k)^2 + (\Delta z_i^k)^2)^{1/2})$ represents the cosine of the polar angle between the neighboring point p_i^k and the center point p_i, $\phi_i^k = (\Delta y_i^k/((\Delta x_i^k)^2 + (\Delta y_i^k)^2)^{1/2})$ denotes the cosine of the azimuthal angle between the neighboring point p_i^k and the center point p_i. The local structure representation in the corresponding prior feature space is as follows:

$$\hat{f}_i^k = (f_i^k - f_i, \|f_i^k - f_i\|). \tag{2}$$

To obtain more comprehensive high-level feature information, the aforementioned geometric space and its corresponding prior feature representations are encoded and aggregated. The specific formula is as follows:

$$\begin{cases} \hat{p}_i = \mathcal{A}\left(\delta\left(\hat{p}_i^k\right)\right), \quad \hat{f}_i = \mathcal{A}\left(\delta\left(\hat{f}_i^k\right)\right), \ l = 1 \\ \hat{p}_i = \mathcal{A}\left(\delta\left(\hat{p}_i^k, F_{\text{out}}^k\right)\right), \quad \hat{f}_i = \mathcal{A}\left(\delta\left(\hat{f}_i^k, F_{\text{out}}^k\right)\right), \ l = 2, 3 \\ c_i^k = \left(\hat{p}_i^k, \hat{f}_i^k, F_{\text{out}}^k\right), \quad l = 4 \end{cases} \tag{3}$$

where δ represents a 1×1 convolution operation designed to increase feature dimensionality for more abstract semantic information. $\mathcal{A}$ denotes the aggregation operation. To maintain network symmetry, summation, averaging, and max pooling are commonly used. In this article, max pooling is used to update the central point feature based on neighboring points. In (3), F_{out}^k represents the high-level abstract semantic feature of the kth neighboring point of the central point p_i (as detailed in Section III-B). It is noteworthy that when layer $= 1$, F_{out}^k is empty, and when layer $= 4$, only LE is performed. Through (3), $\hat{p}_i$ and $\hat{f}_i$ are obtained, which are referred to as high-dimensional geometric semantic information and high-dimensional feature semantic information, respectively.

B. Cross Multiple Anti-Affine Attention

In Section III-A, although LE extracted rich geometric and prior structure features, these two types of features did not interreference. Furthermore, the convolution-based architecture referenced only the neighboring points around the central point for feature updating, disregarding long-range dependencies between central points. Inspired by previous studies [38], this article employs an attention mechanism and introduces a CMAAA module to further extract features from the high-dimensional semantic information obtained in Section III-A, as illustrated in Fig. 2. Here, P and F represent the vectors formed by all the central points $\hat{p}_i$ and $\hat{f}_i$, respectively, as obtained in Section III-A.

Unlike traditional attention mechanisms, this article references [45] and releases the mapping matrix of value (W_v) to reduce the model's parameter count. For the query and key, learnable projection matrices (W_{P_q}, W_{P_k}) and (W_{F_q}, W_{F_k}) are used to map P and F to different feature spaces (P_q, P_k) and (F_q, F_k), respectively. The geometric attention weight coefficient W_P and feature attention weight coefficient W_F can be expressed as

$$W_P = P_q^T P_k, \ W_F = F_q^T F_k. \tag{4}$$

Inspired by Lu et al. [39], we enhanced the weight coefficients W_P and W_F using an anti-affine approach, with the specific formula as follows:

$$\begin{aligned} \hat{W}_P^n &= (\text{softmax}(\text{expand}(\text{maxpooling}(W_P)) + W_P))^n \\ \hat{W}_F^n &= (\text{softmax}(\text{expand}(\text{maxpooling}(W_F)) + W_F))^n \end{aligned} \tag{5}$$

where maxpooling$(\cdot)$ extracts the maximum value of each row of the matrix, while expand$(\cdot)$ repeats the columns to expand them to the original matrix size. The variable n represents the number of repetitions. In (5), the anti-affine operation tends to strengthen the channel expressions with high similarity and suppress the channel expressions with low similarity.

The anti-affine weight coefficients $\hat{W}_P^n$ and $\hat{W}_F^n$ are cross-multiplied with value:F and value:P, respectively. This results in the prior feature $\hat{F}$ being fused with geometric information, and the geometric feature $\hat{P}$ being fused with prior feature information, as expressed in the following equations:

$$\hat{F} = F\hat{W}_P^n, \quad \hat{P} = P\hat{W}_F^n. \tag{6}$$

Considering the differences and importance of $\hat{F}$ and $\hat{P}$, directly adding them together could lead to feature confusion. Therefore, in this article, the two are first concatenated, and a nonlinear layer is then employed to adaptively achieve a deeper cross-fusion of $\hat{F}$ and $\hat{P}$, resulting in features F' and P'. The specific representation is provided below

$$F' = \text{cbl}(\hat{F}, P), \quad P' = \text{cbl}(\hat{P}, F) \tag{7}$$

where cbl represents the combination of 1×1 Convolution, BatchNorm, and LeakyReLU. Finally, the features F' and P' are combined according to the following equation:

$$F_{\text{out}} = \delta(F', P'). \tag{8}$$

It is noteworthy that, to reduce the number of network parameters, through 1×1 convolution, the dimensionality of the output features in (8) is half that of the input features.

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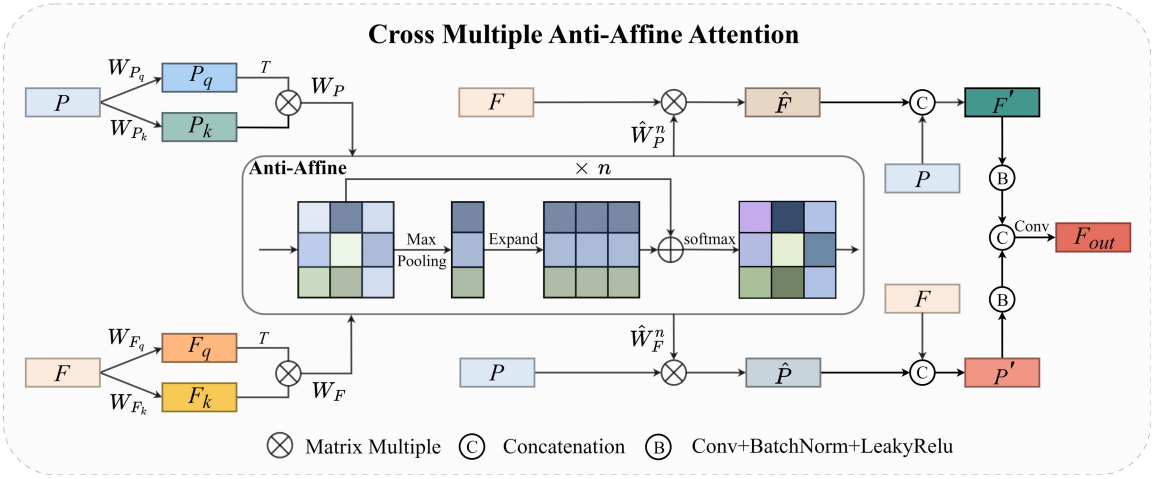


Fig. 2. Architecture of the proposed CMAAA. The correlation coefficient matrix W_P and W_F is generated by multiplying the transposed query with the key, followed by the application of an anti-affine operation resulting in matrices $\hat{W}_P^n$ and $\hat{W}_F^n$, which effectively captures the interchannel correlations.

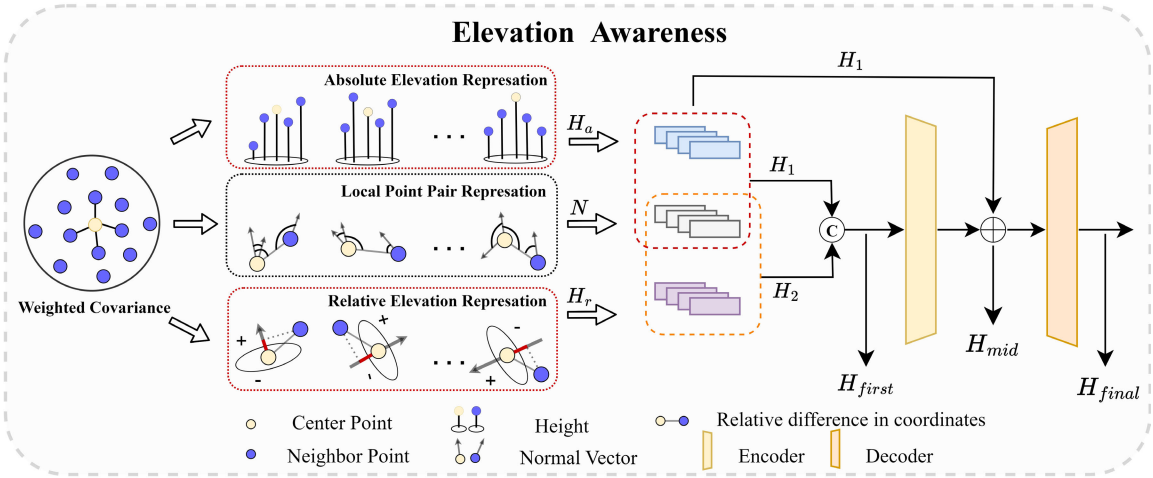


Fig. 3. Architecture of the proposed EA. H_{first} and H_{mid}, as the final outputs, are used for the encoding aggregation of the fourth layer of the encoder and the supplementary input of the first layer of upsampling, respectively.

C. Elevation Awareness

In recent years, elevation information has been extensively used in point cloud segmentation tasks [17], [40], [41], [42]. However, whether elevation information is directly input into the network or used as a self-attention aggregation weight, it only represents a singular treatment. While these approaches clearly layer different categories with significant height differences, they may cause confusion for categories with similar heights. Therefore, this article introduces geometric structure discrimination, point-pair information, and relative elevation to characterize horizontal structural differences, ensuring that elevation information is utilized without bias toward any aspect. The designed EA module is illustrated in Fig. 3.

Surface normals of point clouds are crucial geometric information often used in 3-D reconstruction and object recognition [46]. Typically, true normals are precisely calculated directly from geometric surfaces. However, for point cloud data collected via airborne LiDAR, these are merely discrete sampling points on object surfaces. Therefore, normals obtained using common methods such as PCA [22] and the vector product method [16] usually approximate the true point cloud normals. According to [46], introducing a technique similar to inverse distance weighting (IDW) in the PCA method enhances the contribution of nearby points to surface structures while reducing the influence of distant points, thereby providing more realistic normal and eigenvalue information in sparse point clouds.

Given any point p_i, we use the KNNs algorithm to identify its L neighbors $p_1, p_2, \ldots, p_L$. Based on these neighboring points, the following weighted covariance matrix M_i is constructed:

$$M_i = \sum_{j=1}^{L} \omega_j (p_j - p_i)(p_j - p_i)^T. \tag{9}$$

The formula for calculating the weighted coefficient ω_j is as follows:

$$\omega_j = \frac{\max |p_j - p_i| - |p_j - p_i|}{\sum_{j=1}^{L} \max |p_j - p_i| - |p_j - p_i|} \tag{10}$$

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where $\max|p_j - p_i|$ denotes the maximum distance from p_i to all of the L neighboring points. Subsequently, the eigenvector corresponding to the smallest eigenvalue $\lambda_{\min}$ of the weighted covariance matrix M_i serves as an approximate estimate of the true normal vector of p_i. For the convenience of subsequent model construction, this normal vector is normalized and denoted as n_i.

The three largest eigenvalues $(\lambda_{i_1}, \lambda_{i_2}, \lambda_{i_3})$ of the weighted covariance matrix M_i are selected, and any value less than 0 truncated to 0 and then normalized so that their sum is 1. In reference to [17], these eigenvalues are used to compute complex shape descriptors such as linearity, planarity, sphericity, and so forth. In contrast to [17], the vector composed of these three eigenvalues is used as a representation of the geometric structure information of the point. Thus, it is reasonable to assume that objects with similar geometric structure information are more likely to belong to the same category, so similar points should be given higher weights during aggregation. Define $d\lambda_j = 1/[(\lambda_{j_1} - \lambda_{i_1})^2 + (\lambda_{j_2} - \lambda_{i_2})^2 + (\lambda_{j_3} - \lambda_{i_3})^2 + \varepsilon]$, where $(\lambda_{j_1}, \lambda_{j_2}, \lambda_{j_3})$ represent the three largest eigenvalues of the neighboring point p_j, and ε is a small, positive value, usually 10^{-4}. Let $\omega\lambda_j = d\lambda_j / \sum_{j=1}^{L} d\lambda_j$, then the geometric similarity weight of neighboring points $\{p_1, p_2, \ldots, p_L\}$ relative to the central point p_i is calculated as follows:

$$\omega\lambda = [\omega\lambda_1, \omega\lambda_2, \ldots, \omega\lambda_L]. \tag{11}$$

1) Absolute Elevation Representation: In constructing absolute elevation information for each point, we adopted the concept of graph convolution to learn elevation features using local information by considering the L neighboring points around the central point. The original height vector of the L neighboring points, obtained using the KNN method, is represented as $\mathbf{z} = (z_1, z_2, \ldots, z_L)$. Let $Z = \mathbf{z} \cdot \omega\lambda$ denote the absolute elevation vector, incorporating a geometric similarity discriminant. Given the unique characteristics of elevation information, we employed various aggregation methods to enhance the representation of absolute elevation. The specific formulation for absolute elevation H_a is as follows:

$$H_a = \text{concat}(\max(Z) - \min(Z), \text{mean}(Z), \text{std}(Z)) \tag{12}$$

where $\max(Z) - \min(Z)$ represents the range of height fluctuations, $\text{mean}(Z)$ indicates the overall height level of the local region, and $\text{std}(Z)$ represents the dispersion relative to the mean height.

2) Relative Elevation Representation: To ensure that the elevation information of the neighboring point p_j relative to the point p_i remains invariant to translation and rotation, we use a stable local reference frame based on the normal vector instead of the Cartesian coordinate system. Since the normal vector n_i is a unit vector, the relative elevation information (the projection length of the relative coordinate e_{ij} from the neighboring point to the central point on the normal vector n_i) can be expressed as the inner product of e_{ij} and n_i, denoted as $z_j^p = \text{dot}(e_{ij}, n_i)$. The relative elevation vector composed of the neighboring points is then expressed as $\mathbf{z}^p = (z_1^p, z_2^p, \ldots, z_L^p)$. Let $Z^p = \mathbf{z}^p \cdot \omega\lambda$. Then, the relative elevation feature H_r with geometric similarity discriminant is expressed as

$$H_r = \text{concat}(\max(Z^p), \min(Z^p), \text{mean}(Z^p), \text{std}(Z^p)). \tag{13}$$

It is worth noting that due to the presence of positive and negative projections, in (13), unlike the aggregation method in (12), we separate $\max(Z^p) - \min(Z^p)$ into $\max(Z^p)$ and $\min(Z^p)$.

3) Local Point Pair Expression: To enrich the horizontal expression of elevation information, and drawing on the literature [22], a new local expression is constructed for each neighboring point based on the normal vector n_i of the point p_i, denoted as $r_j = (|e_{ij}|, \theta(n_i, e_{ij}), \theta(n_j, e_{ij}), \theta(n_i, n_j))$. Here, $|e_{ij}|$ denotes the magnitude of the relative coordinates e_{ij} of the neighboring point p_j with respect to the point p_i; $\theta(n_i, e_{ij})$ and $\theta(n_j, e_{ij})$ represent the angles between the normal vectors n_i and n_j of the respective points p_i and p_j and their relative coordinates; Finally, $\theta(n_i, n_j)$ indicates the angle between the normal vector n_i and the normal vector of the neighboring point p_j. Let $N_i = (r_1, r_2, \ldots, r_L)$. The formula for expressing local point pair information is as follows:

$$N = \text{concat}(\text{mean}(N_i) + \max(N_i), \text{std}(N_i)). \tag{14}$$

4) Relative-Absolute Elevation With Point Pair Information: To improve the localization of neighboring points relative to the centroid and enhance structural variability horizontally, this article adopts a self-coding framework to integrate absolute-relative elevation information (high dimensionality $\rightarrow$ low latitude $\rightarrow$ high latitude). This self-coding approach symmetrically aligns with the downsampling-upsampling network based on the PointNet++ architecture (low dimensionality $\rightarrow$ high dimensionality $\rightarrow$ low latitude), enabling the network to learn richer features. Let $H_1 = \text{concat}(H_a, N)$, $H_2 = \text{concat}(H_r, N)$, and $H_{\text{first}} = (H_1, H_2)$. To further enrich the expression of absolute elevation, the encoding phase can be expressed as

$$H_{\text{mid}} = \text{MLP}(H_{\text{first}}) + H_1. \tag{15}$$

The decoding phase can be expressed as

$$H_{\text{final}} = \text{MLP}(H_{\text{mid}}). \tag{16}$$

The mean squared error is used as the loss function to adjust the elevation information expression adaptively through the network

$$\text{loss1} = \| H_{\text{final}} - H_{\text{first}} \|_2^2 . \tag{17}$$

The relative-absolute elevation expression incorporating point pair information H_{mid} is obtained using (17).

Next, to fully utilize the elevation information to enhance the network's expression capability, the obtained H_{first} and H_{mid} are used for the encoding aggregation of the fourth layer of the encoder and the supplementary input of the first layer of upsampling, respectively, as shown in Fig. 4. The specific formula is as follows:

$$F_{\text{out}}^i = \left(\mathcal{A}\left(\delta\left(c_i^k, H_{\text{first}}^i\right)\right), H_{\text{mid}}^i\right) \tag{18}$$

where H_{first}^i and H_{mid}^i denote the elevation information H_{first} and H_{mid} for the ith central point in the fourth layer (l_4) of

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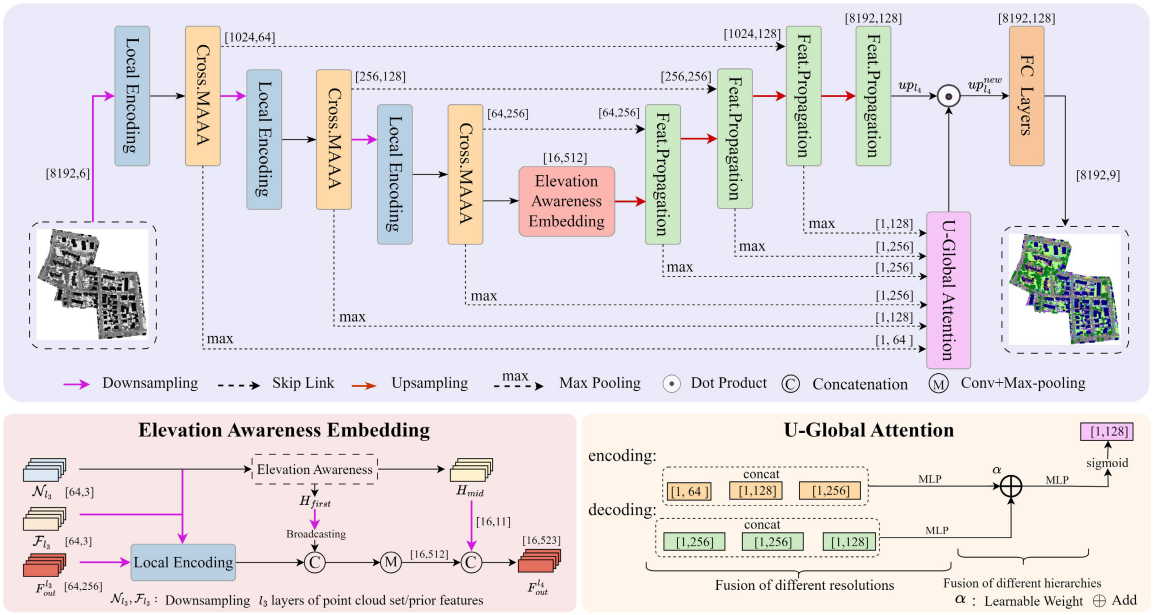


Fig. 4. Overall model architecture. The proposed DAEA-Net is composed of LE, Cross Multiple Anti-Affine Attention (abbreviated as Cross.MAAA in the figure), EA Embedding (consisting of LE as described in (3) when $l = 4$ and EA as described in Section III-C), UGA, feature propagation (composed of concatenation and MLP blocks, abbreviated as Feat.Propagation in the figure), and fully connected layers (abbreviated as FC Layers in the figure). Among these, EA Embedding and UGA are described in detail.

the downsampling process, respectively. F^i_{out} is the output of the EA Embedding and also the input of the first layer l_1 of the upsampling (as shown in Fig. 4).

D. U-Global Attention

Many network models employ a pyramid-like architecture with skip connections [17], [18], [19], [20] to facilitate the interaction between low-level and high-level features. However, these operations are complex and involve a substantial number of parameters. To address this issue, we introduce a lightweight UGA module in this study, which facilitates the cascade between encoding and decoding levels, thereby enhancing the expressiveness of downstream classifiers, as depicted in Fig. 4. The UGA module is described as follows:

$$\text{up}^{\text{new}}_{l_4} = \text{up}_{l_4} \cdot \text{sigmoid}\big(\text{MLP}\big(\alpha \cdot d.n_{\text{all}} + \text{up}_{\text{all}}\big)\big) \tag{19}$$

where $\text{up}^{\text{new}}_{l_4}$ represents the final output of the upsampled l_4 layer and serves as the input for downstream tasks, while up_{l_4} is the output of the upsampled l_4 layer. The parameter α is a learnable weighting coefficient that adaptively determines the importance of upsampling and downsampling. In (19), $d.n_{\text{all}}$ and up_{all} represent the cascade from low-level to high-level features during downsampling and upsampling, respectively, as expressed in the following mathematical formulas:

$$\begin{cases} d.n_{\text{all}} = \text{MLP}\big(\max(d.n_{l_1}), \max(d.n_{l_2}), \max(d.n_{l_3})\big) \\ \text{up}_{\text{all}} = \text{MLP}\big(\max(up_{l_1}), \max(\text{up}_{l_2}), \max(\text{up}_{l_3})\big) \end{cases} \tag{20}$$

where the output dimensions of $d.n_{\text{all}}$ and up_{all} are each 128. The global point representations of the first three downsampling layers l_1, l_2, and l_3 are $\max(d.n_{l_1})$, $\max(d.n_{l_2})$, and $\max(d.n_{l_3})$, with corresponding dimensions $1 \times d_1$, $1 \times d_2$, and $1 \times d_3$, respectively. The global point representations of the first three upsampling layers l_1, l_2, and l_3 are $\max(up_{l_1})$, $\max(up_{l_2})$, and $\max(up_{l_3})$, with corresponding dimensions $1 \times d_4$, $1 \times d_5$, and $1 \times d_6$, respectively.

E. Network Architecture

The proposed network architecture is illustrated in Fig. 4. During the encoding phase, the point cloud is downsampled progressively via FPS, reducing the point count to 1024, 256, 64, and 16, while corresponding feature dimensions are 64, 128, 256, and 512. A LE module is applied in each downsampling layer to construct and encode the local structure of different features. To overcome the receptive field limitations of convolutional architectures, a CMAAA module is applied to the first three layers for long-range feature learning, enabling the network to understand relationships between neighborhoods. In the final downsampling layer, an EA module is introduced to enhance feature representation in the vertical direction. During upsampling, IDW is applied four times consecutively to restore the point count to 64, 256, 1024, and 8192, with corresponding feature dimensions of 256, 256, 128, and 128. Skip connections between each pair of downsampling and upsampling layers fuse different information levels at the same resolution. Additionally, a UGA module is used to capture comprehensive global information, serving as attention weights to enhance feature representation for downstream classifier inputs. Finally, a MLP serves as a downstream classifier, predicting semantic labels for each point.

IV. Experiment

To validate the effectiveness of the proposed DAEA-Net, we conducted experiments on two datasets, ISPRS [47] and

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TABLE I
CLASS DISTRIBUTION OF THE ISPRS DATASET

Class	Training Set	Test Set
Power	546	600
Low_veg	180,850	98,690
Imp_ surf	193,723	101,986
Car	4,614	3,708
Fence	12,070	7,422
Roof	152,045	109,048
Facade	27,250	11,224
Shrub	47,605	24,818
Tree	135,173	54,226
Total	753,876	411,722

TABLE II
CLASS DISTRIBUTION OF THE GML DATASET

Class	Training Set	Test Set
Ground	557,142	439,989
Building	98,244	19,592
Car	1,833	3,235
Tree	381,677	531,852
Low_veg	35,093	7,758
Sum	1,073,989	1,002,426

GML [48]. We first introduce the datasets and their preprocessing, followed by a description of the evaluation metrics and experimental setup. Next, we present the experimental results, and finally, we perform ablation studies on the ISPRS dataset.

A. Datasets

ISPRS dataset: This dataset contains airborne laser scanning (ALS) point clouds from three areas in Stuttgart, Germany. The data were collected at approximately 500 m altitude with a 45° flight angle. Each region includes nine categories: powerline (power), low vegetation (low_veg), impervious surface (imp_surf), car, fence, roof, facade, shrub, and tree. The dataset is split into training (753 876 points) and testing (411 722 points) sets, with each point cloud containing x, y, z, intensity, number of echoes, total number of echoes. The category distribution is shown in Table I. Based on the multispectral orthophoto provided, color information for each point is available [41]. The training and testing areas are divided into 30 × 30 blocks. During training, each block samples 8192 points, while during testing, all points in each block are used. Additionally, we apply a random voting method to obtain the prediction for each point, as detailed in [49]. Data augmentation during training includes random z-axis rotation, scaling, and random wrapping of xyz.

GML dataset: This dataset, provided by the Graphics and Media Laboratory of Moscow State University, covers a wide range of areas with five categories: ground, building, car, tree, and low vegetation (low_veg). The training set has 1 073 989 points, and the test set has 1 002 426 points, with each point containing only xyz coordinates. The category distribution is shown in Table II. Due to the sparse density, each training block samples 4096 points, while the test set uses all points in each block, as detailed in [49]. The preprocessing is similar to the ISPRS dataset, and three normalizations (Min–Max, Z-Score, Decimal Scaling) are used as prior features.

B. Configuration

Experimental Configuration: All experiments were conducted on an NVIDIA GeForce RTX 4090 24-GB GPU within a PyTorch 1.10.0 + CUDA 11.3 framework. During the training phase, we set the batch size to 6, the maximum number of epochs to 500, the initial momentum to 0.1, and the initial learning rate to 0.002 using the AdamW optimizer [50]. Every 100 epochs, the learning rate decays to 25% of its previous value, and the momentum decays to 50% of its previous value. A dropout rate of 0.5 is applied in the penultimate fully connected layer. On the ISPRS dataset, the total training time is 1.4 h with a memory usage of 7 GB; on the GML dataset, the total training time is 4 h with a memory usage of 5.6 GB.

Evaluation Metrics: We use two metrics to evaluate our network model comprehensively: overall accuracy (OA) and $F1$ Score ($F1$). OA measures the classification performance across all categories, calculated as the ratio of correctly classified points to the total number of points tested. The $F1$ Score, calculated for each category, considers both the precision and recall of the model, making it suitable for evaluating performance in imbalanced datasets. The mathematical definitions of these metrics are as follows:

$$\begin{cases} \text{Precision} = \dfrac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Recall} = \dfrac{\text{TP}}{\text{TP}+\text{FN}} \\ F1\text{Score} = 2 \times \dfrac{\text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}} \\ \text{OA} = \dfrac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}. \end{cases} \tag{21}$$

The performance metrics used include true positive (TP), true negative (TN), false positive (FP), and false negative (FN). The mean $F1$ score (m$F1$) is calculated by averaging the $F1$ scores across all categories.

Loss Function: To optimize the network model parameters and address the extreme class imbalance in the data, more attention is required for underrepresented categories. A common approach is to use a weighted loss function, where each category's weighting coefficient is calculated as the inverse ratio of the number of points in that category to the total number of points. The formula is as follows:

$$W_c = \frac{1}{\ln\left(\alpha_1 + \frac{N_c}{\sum_{c=1}^{C} N_c}\right)} \tag{22}$$

where α_1 is the balance coefficient, set to 1.2 [49]. In this article, C represents the total number of categories, and N_c denotes the number of points in the cth category. The loss function employed in this study is expressed as

$$\text{Loss} = \text{loss2} + \beta \cdot \text{loss1} \tag{23}$$

where β denotes the weighting factor, set to $\beta = 1$ for the ISPRS dataset and $\beta = 0.5$ for the GML dataset. Referring

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TABLE III

CONFUSION MATRIX OF THE PROPOSED NETWORK ON THE ISPRS BENCHMARK DATASET. PRECISION (%), RECALL (%), AND $F1$ SCORE (%) ARE REPORTED FOR EACH CATEGORY. THE OA IS 85.31% AND THE MEAN $F1$ SCORE IS 74.71%

Category	Power	Low_veg	Imp_surf	Car	Fence	Roof	Facade	Shrub	Tree	Precision	Recall	F1 score
Power	**475**	2	0	0	0	75	4	0	44	75.3	79.2	77.17
Low_veg	0	**80,549**	10,277	146	751	1,131	290	3,527	2,019	83.5	81.6	82.53
Imp_surf	0	5,600	**95,626**	36	31	363	32	290	8	89.5	93.8	91.60
Car	0	152	93	**2,896**	40	76	18	419	14	87.1	78.1	82.37
Fence	0	1,011	164	37	**3,163**	121	54	1,961	911	51.6	42.6	46.66
Roof	135	1,730	248	2	26	**103,745**	1,104	1,143	915	95.2	95.1	95.19
Facade	10	982	146	74	10	1,576	**6,651**	1,156	619	75.6	59.3	66.43
Shrub	5	5,065	238	126	1,336	1,120	330	**11,116**	5,482	47.2	44.8	45.96
Tree	6	1,420	21	7	778	723	318	3,942	**47,011**	82.4	86.7	84.51

to the literature [50], loss2 represents the weighted label smoothing loss function, expressed as follows:

$$\text{loss2} = (1-\gamma)\sum_{c=1}^{C} W_c \times (y_c \log \hat{y}_c + (1-y_c)\log(1-\hat{y}_c)) + \gamma \cdot \left(-\frac{1}{C}\sum_{c=1}^{C}\log(y_c)\right) \tag{24}$$

where y_c denotes the one-hot vector of the actual label, while $\hat{y}_c$ represents the probability vector of the cth sample being predicted as the cth category. γ indicates the label smoothing coefficient, set to $\gamma = 0.2$ in this study.

C. Semantic Segmentation Results

1) Results of the ISPRS Data Experiment: To quantitatively evaluate the classification performance of our proposed model on the ISPRS dataset, Table III presents the confusion matrix. As shown in Table III, the model effectively recognizes most categories. Six of the nine categories have $F1$ scores around 80%, including powerline, low vegetation, impervious surfaces, cars, roofs, and trees. However, a small percentage of facades were misclassified as roofs, shrubs, and low vegetation. This misclassification primarily occurs because facades have a wider range of heights and are spatially close to roofs, shrubs, and low vegetation, leading to confusion. Fences were misclassified as shrubs in nearly 26% of the points, largely due to significant overlap between these two categories. Additionally, shrubs had a low $F1$ score (45.96%), and were frequently misclassified as low vegetation and trees, mainly due to the extreme similarity in the shape patterns of shrubs, low vegetation, and trees in the horizontal direction. Notably, our model distinguishes powerline from highly similar and alternating roofs, despite the small sample size. Similarly, cars, although having fewer and more widely dispersed points, are recognized effectively. To present our classification performance more clearly, Figs. 5 and 6 illustrate the classification and error graphs, respectively. Fig. 5 shows that most test set points are correctly classified, while Fig. 6 reveals that misclassified points are concentrated at object junctions and in regions with complex scenes.

To further validate our proposed method, we compared it with several mainstream approaches, including PointNet++, PointSIFT, RandLA-Net, KPConv, D-FCN, DANCE-NET, VD-LAB, RFFFS-Net, *DA-Net, *GAHD-Net, *GraNet, LGENet, and *Dense-LGEANet (where * indicates methods that use elevation information). Table IV presents the specific

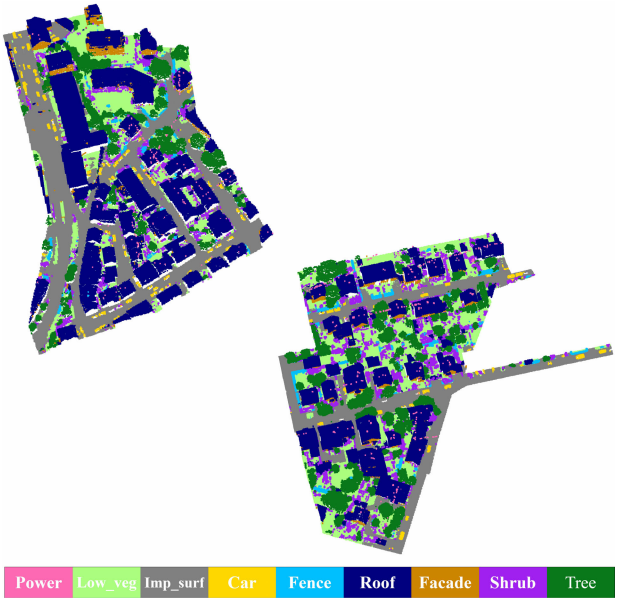


Fig. 5. Classification results of the proposed DAEA-Net on the ISPRS benchmark dataset. Colors represent different categories. The point categories in each scene are presented in the colors of the legend at the bottom.

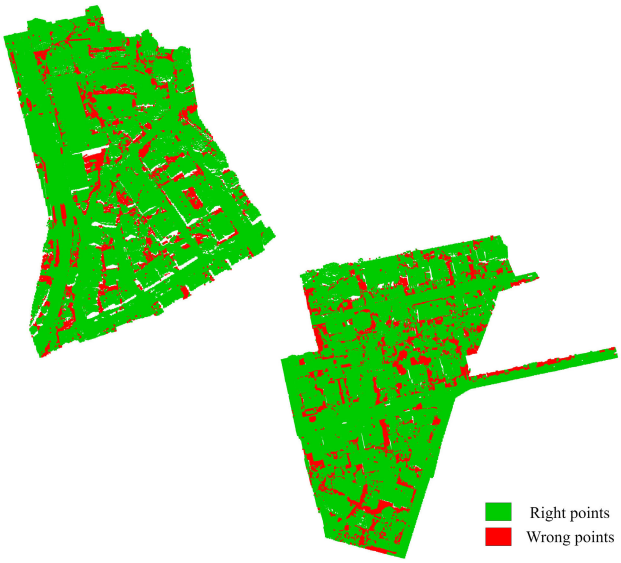


Fig. 6. Error map of the proposed DAEA-Net on the ISPRS benchmark dataset. The green points and red points represent the correct and wrong segmentation results, respectively.

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TABLE IV

QUANTITATIVE RESULTS OF THE DIFFERENT METHODS ON THE ISPRS DATASET. RESULTS INCLUDE $F1$ SCORE (%), OA (%), AND m$F1$ SCORE (%) FOR EACH CATEGORY. IN THE COLUMNS, BOLD FONT DENOTES THE HIGHEST RESULT, UNDERLINING DENOTES THE SECOND BEST RESULT, DOUBLE UNDERLINING DENOTES THE THIRD BEST RESULT, AND * DENOTES NETWORKS WITH ELEVATION INFORMATION

Method	Power	Low_veg	Imp_surf	Car	Fence	Roof	Facade	Shrub	Tree	OA	mF1
PointNet++ [33]	67.60	80.93	91.65	76.84	35.08	91.36	57.02	41.59	77.57	81.99	68.85
PointSIFT [51]	55.70	80.70	90.90	77.80	30.50	92.50	56.90	44.40	79.60	82.20	67.70
RandLA-Net [36]	68.80	82.10	91.30	76.60	43.80	91.10	61.90	45.20	77.40	82.10	70.90
KPConv [35]	73.50	78.70	88.80	79.40	33.00	94.20	61.30	45.70	82.00	81.70	70.60
D-FCN [52]	70.40	80.20	91.40	78.10	37.00	93.00	60.50	46.00	79.40	82.20	70.70
DANCE-NET [53]	68.40	81.60	92.80	77.20	38.60	93.90	60.20	47.20	81.40	83.90	71.20
VD-LAB [49]	69.30	80.50	90.40	79.40	38.30	89.50	59.70	45.70	77.20	81.40	70.20
RFFFS-Net [18]	75.50	80.00	90.50	78.50	45.50	92.70	57.90	48.30	75.70	82.10	71.60
*DA-Net [41]	69.70	**84.30**	**93.00**	78.80	37.40	**95.50**	61.80	46.60	83.70	**85.90**	72.30
*GAHD-Net [17]	75.40	82.00	91.60	77.80	44.20	94.40	61.50	49.60	82.60	84.50	73.20
*GraNet [40]	67.70	82.70	91.70	80.90	**51.10**	94.50	62.00	**49.90**	82.00	84.50	73.60
LGENet [54]	76.50	82.10	91.80	80.00	40.60	93.80	64.70	**49.90**	83.60	84.50	73.70
*DenseLGEANet [42]	72.20	80.80	92.10	78.70	41.20	93.90	61.50	47.20	80.50	83.20	72.00
DAEA-Net (Ours)	**77.17**	82.53	91.60	**82.37**	46.66	95.19	**66.43**	45.96	**84.51**	85.31	**74.71**

results. It is evident that our proposed method outperforms the others, achieving the highest m$F1$ score of 74.71%. Compared with PointNet++ as the benchmark, our network improved the OA and m$F1$ values by 3.32% and 5.86%, respectively.

Compared to LGENet, a KPConv-based network, our method improved the OA and m$F1$ values by 0.81% and 1.01%, respectively, clearly demonstrating that our network can achieve good results without relying on distinctive kernel point convolutions. Compared to elevation information-based networks like DA-Net, GAHD-Net, GraNet, and DenseLGEANet, our method achieved the best metrics for objects with significantly different height distributions (e.g., trees, fences, and low vegetation). This clearly shows that our EA module improves the vertical differentiation of objects.

For objects with similar and adjacent height distributions (e.g., powerline, trees, and roofs), their m$F1$ values improved to 77.17%, 84.51%, and 95.19% (second-best), respectively. This demonstrates that our network successfully incorporated horizontal structural information into height data, enhancing the representation of elevation features.

Additionally, our method has achieved good results in challenging categories such as powerlines, cars, and facades. This is primarily because the proposed method can facilitate mutual learning between features in different neighborhoods within the same layer as well as between different layers. Consequently, it can capture local and global features at various scales, thereby enhancing the long-range learning capability for classes with limited labels.

2) Results of the GML Data Experiment: To demonstrate the transferability of our method, we conducted experiments on the GML dataset and compared it with other methods, including PointNet++, PointCNN, DensePoint, RS-CNN, DGCNN, PointConv, KPConv, GACNet, ConvPoint, and VD-LAB. The specific results are presented in Table V. Unlike the ISPRS dataset, the GML scenario suffers from extreme class imbalance, so greater attention is given to class $F1$ score and the overall m$F1$ score when evaluating model performance.

TABLE V

QUANTITATIVE RESULTS OF THE DIFFERENT METHODS ON THE GML DATASET. RESULTS INCLUDE $F1$ SCORE (%), OA (%), AND m$F1$ SCORE (%) FOR EACH CATEGORY. IN THIS COLUMN, BOLD FONT INDICATES THE HIGHEST RESULTS AND UNDERLINING INDICATES THE SECOND BEST RESULTS

Method	Ground	Building	Car	Tree	Low_veg	OA	mF1
PointNet++ [33]	83.83	27.38	39.23	93.99	15.55	85.28	52.00
PointCNN [34]	92.32	42.09	22.94	94.64	22.38	91.99	55.07
DensePoint [55]	94.35	51.25	32.67	96.71	8.36	94.12	56.67
RS-CNN [56]	92.80	35.01	35.24	96.15	15.06	93.27	54.85
DGCNN [12]	95.33	46.88	27.02	97.79	11.40	94.67	55.68
PointConv [15]	95.76	53.14	45.98	97.35	28.92	95.11	64.23
KPConv [35]	88.65	43.24	22.76	96.15	11.01	90.02	52.36
GACNet [57]	96.34	77.73	0.00	97.98	34.75	95.85	61.36
ConvPoint [58]	**96.54**	80.39	24.85	96.02	15.10	95.05	62.58
VD-LAB [49]	95.38	60.85	41.80	97.95	35.56	95.49	66.31
DAEA-Net (Ours)	96.51	**81.41**	**59.66**	**98.08**	**46.50**	**96.50**	**76.43**

Our method achieved the best classification performance, significantly outperforming other methods in terms of OA and m$F1$ metrics, reaching 96.50% and 76.43%, respectively. Additionally, our network exhibited excellent performance across all categories. Compared to PointNet++, KPConv introduces kernel point convolution, enabling the network to better capture object contours and enhance its overall discriminative power for complex objects. However, the $F1$ score of KPConv in the car category was only 22.76%, which is 16.47% lower than PointNet++, indicating that simply enhancing geometric perception on the GML dataset does not improve the network's discriminative performance. In contrast, VD-LAB reduces the complexity of feature extraction through decoupling while achieving better results, with an $F1$ score of 41.80% for the car category. Therefore, designing an effective feature extraction module for the GML dataset is a challenging task. The PointConv network enhances feature characterization by introducing point density to differentiate between points, resulting in an $F1$ score of 45.98% for the

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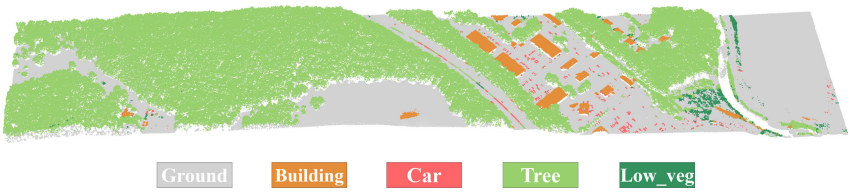


Fig. 7. Error map of the proposed DAEA-Net on the GML benchmark dataset. Colors represent different categories.

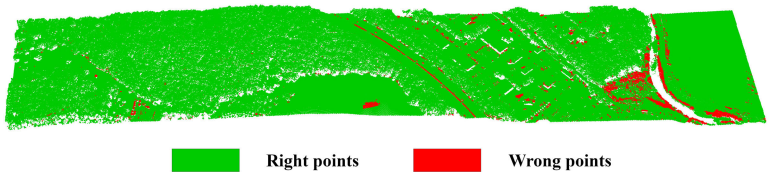


Fig. 8. Error map of the proposed DAEA-Net on the GML benchmark dataset. The green points and red points represent the correct and wrong segmentation results, respectively.

car category. Compared to PointConv, the $F1$ score of our proposed method in the car category improved by 13.68%, which can be attributed to our CMAAA and UGA modules that expand the network's receptive field. This allows the network to recognize the same objects distributed across different regions. Additionally, compared to other methods, our method showed significant improvement in adjacent building and tree recognition, demonstrating that our elevation module enhances differentiation horizontally.

To provide a more intuitive understanding of the classification performance of the proposed method, Fig. 7 shows the classification results on the GML dataset, while Fig. 8 illustrates the classification errors. As shown in these figures, very few points are misclassified, and the results closely reflect the real distribution. Furthermore, Fig. 9 compares the classification results of the benchmark network PointConv and our method across three complex sections. The proposed method performs better, as highlighted in the red circles.

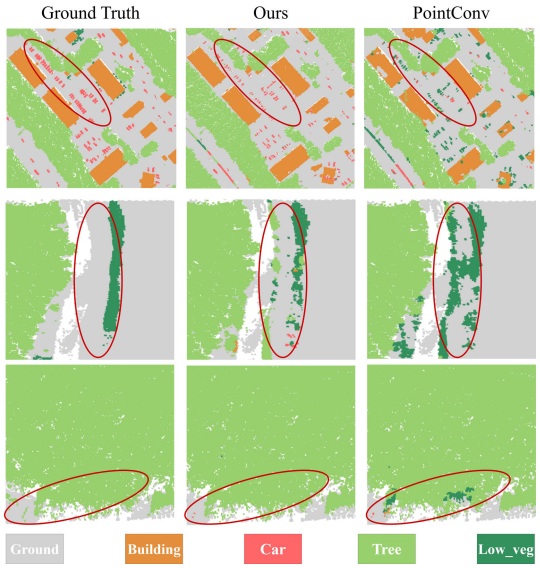


Fig. 9. Comparison of small graphs between the proposed DAEA-Net and PointConv on the GML benchmark dataset. The area marked by the black ellipse is where our method performs better than other methods.

V. Discussion

In this section, we explore in detail the validity of each module constructed in this article and the design ideas behind these modules. All experiments are presented on the ISPRS benchmark dataset.

A. Ablation Studies of Core Components

Our network framework consists of three main components: CMAAA, EA, and UGA. To evaluate the performance of these components, ablation experiments were conducted, as shown in Table VI. Specifically, Model A adds two-channel structural coding to the baseline PointNet++ network [see (3)], utilizing the $F_{\text{out}} = \text{MLP}(P + F)$ approach for feature fusion with a label-smoothing loss function. Model G represents our proposed DAEA-Net, while Model C, Model D, and Model E represent the DAEA-Net without UGA, CMAAA, and EA, respectively. Model F replaces the label-smoothing loss function in DAEA-Net with a weighted cross-entropy loss function [49].

As shown in Table VI, Model G improved OA and m$F1$ values by 0.57% and 0.54%, respectively, compared to Model F. This indicates that incorporating a smoothing strategy in the weighted cross-entropy loss function effectively reduced overfitting to noise and anomalous labels in the training data, thereby improving the model's generalization ability. The OA and m$F1$ values of Model C are lower than those of Model G, suggesting that the proposed UGA module facilitates the interaction characterization of global information at different scales. Model E lacks the elevation information representation compared to Model B, and its OA and m$F1$ values significantly decrease, showing that the elevation module enhances category differentiation both vertically and horizontally. Models C, D, and E showed decreases in m$F1$ values of 1.08%, 1.61%, and 0.98%, respectively, compared to our network, highlighting the importance of the CMAAA module. This module employs a special attentional mechanism to learn converging centroids through cross-fertilization, effectively fusing

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TABLE VI

ABLATION ANALYSIS OF DIFFERENT MODULES ON THE ISPRS DATASET. RESULTS INCLUDE OA (%), m$F1$ SCORE (%), MODEL PARAMETERS (PARAMS) (IN MILLIONS), FLOPS (GIGA), PER-EPOCH TIME (INCLUDING TRAINING AND VALIDATION) (S), AVERAGE INFERENCE TIME (AVG. INFERENCE TIME, AVERAGED OVER TEN VOTES) (S), AND INFERENCE THROUGHPUT (SAMPLES PER SECOND, SPS) (SAMPLES PER SECOND, SPS) FOR EACH COMBINATION

Model	CMAAA	EA	UGA	Loss2	OA (%)	mF1 (%)	Params (M)	FLOPs (G)	Per-Epoch Time (s)	Avg. Inference Time (s)	Inference Throughput (SPS)
A				✓	83.45	71.15	2.47	3.054	10.56	9.943	7.946
B	✓			✓	84.26	72.97	3.85	3.257	10.71	9.985	7.912
C	✓	✓		✓	84.43	73.63	3.87	3.259	10.96	10.024	7.881
D		✓	✓	✓	84.58	73.10	2.70	3.056	10.83	10.008	7.894
E	✓		✓	✓	84.63	73.73	4.07	3.257	10.92	10.013	7.889
F	✓	✓	✓		84.74	74.17	4.08	3.260	11.12	10.115	7.810
G	✓	✓	✓	✓	**85.31**	**74.71**	4.08	3.260	11.12	10.115	7.810

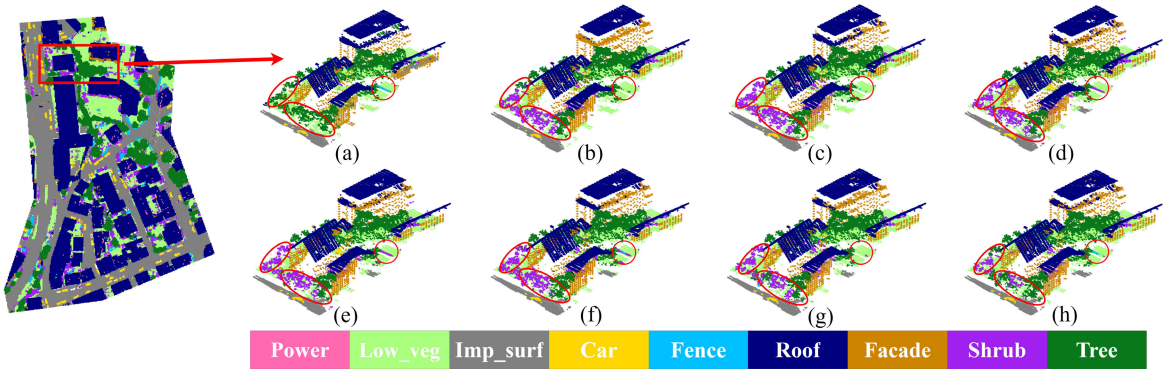


Fig. 10. Categorized maps of locally magnified areas under different models on the ISPRS dataset. (a) Ground truth. (b) Model A. (c) Model B. (d) Model C. (e) Model D. (f) Model E. (g) Model F. (h) Model G.

information of different types while enhancing the network's receptive field and long-range dependency among neighbors.

To further investigate the CMAAA module's impact, Model B was introduced—a separate CMAAA module on top of Model A. The results show that the OA and m$F1$ values of Model B improved by 0.81% and 1.82%, respectively, compared to Model A. Its m$F1$ score is only 0.13% lower than that of Model D, which contains the other two modules, further demonstrating the significant efficacy of the CMAAA module in feature enhancement. By comparing the progressively higher OA and $F1$ scores of Models B, C, and G, as well as Models B, E, and G, these three modules collectively contribute to the effectiveness of network learning from different perspectives. From the other evaluation metrics in Table VI, it is evident that the introduction of the EA and UGA modules has a negligible impact on the increase in floating point operations (FLOPs). Additionally, the training time per epoch, average inference time, and throughput of all models are very similar. However, the average inference time is nearly 10 s, which is critical for real-time inference tasks. This is mainly because we perform inference on a single GPU, processing one batch at a time. Using multiple GPUs in parallel could significantly reduce this time. Furthermore, the introduction of the CMAAA module increases the network parameters by nearly 1.2 million, which is 75 times that of the EA module (approximately 0.016 million) and six times that of the UGA module (approximately 0.2 million). Therefore, in in Sections V-B and V-C, we will further explore the design of the CMAAA module, which has the best performance, and the EA module, which has the fewest parameters but slightly lower performance. Fig. 10 shows a categorized map of locally magnified regions under different models on the ISPRS dataset. In the three sections highlighted in the figure, our method provides better classification results that closely resemble the true values.

B. Ablation of CMAAA

1) Ablation Studies on How to Combine: We are particularly interested in three aspects of this module's design: 1) the necessity of the mapping matrix of value; 2) the methods for fusing different types of features; and 3) the derivation of attention weights. All experiments in this section are conducted on Model G, as detailed in Table VI and shown in Table VII. A_1 indicates that the mapping matrix W_v is essential in the attention design. A_2 and A_3 represent the operations of summation fusion and postconcatenation fusion of the original inputs in (7), respectively [e.g., $F' = \hat{F} + P$ and $F' = \text{cbl}(\hat{F}, P)$]. A_4 and A_5 illustrate the operations of summation fusion and postconcatenation fusion in (8), respectively [e.g., $F_{\text{out}} = F' + P'$ and $F_{\text{out}} = \text{conv}(F', P')$]. A_6 shows that query and key multiplication is used to compute weight coefficients, while A_7 and A_8 indicate that weight coefficients are computed through affine and anti-affine transformations, respectively, based on A_6. Table VII demonstrates that in attention design, the same effect can be achieved without the matrix mapping on VALUE, aligning with the concept shown in [45]. The results from Models B, C, and our method reveal that the additive fusion approach might not

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TABLE VII
ABLATION STUDY OF CMAAA IN DIFFERENT COMBINATIONS OF MODALITIES

Model	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	OA	mF1
Ours			✓		✓			✓	**84.26**	**72.97**
A	✓		✓		✓			✓	84.15	72.82
B		✓			✓			✓	83.92	72.19
C			✓	✓				✓	84.01	72.48
D			✓		✓		✓		83.26	71.52
E			✓		✓	✓			83.63	71.85

fully account for the complementarity and significance of the differences between the two features, leading to performance degradation. Conversely, the concatenation and fully connected layer approach leverages the network's adaptive learning to more effectively fuse these two distinct feature types, thereby enhancing global contextual information.

The comparison of Models D, E, and our method reveals that the anti-affine design amplifies the expression of self-attention weights, promoting an optimal model configuration. Unlike the subtraction operation used to filter noise in indoor point clouds [39], the addition operation applied to large-scale outdoor datasets compensates for the lack of attention expression.

2) Number of Anti-Affine Operations: The anti-affine operation in Table VII was used consecutively twice because a single anti-affine may not fully capture the similarity between channels. To better understand the effect of this operation, we conducted experiments on the final network model with three different numbers of anti-affine operations ($n = 1, n = 2$, and $n = 3$), with the results presented in Table VIII. Table VIII shows that the model performs best when $n = 2$. For classes of objects that are widely dispersed spatially (e.g., powerline, cars, fences, and facades), there is a slight improvement in performance compared to $n = 1$. This suggests that increasing the number of anti-affine operations aids the learning of interactions between different neighborhoods, enhancing the network's long-range dependence.

However, the misclassification rate of structurally similar objects (e.g., shrubs and trees) increased when the number of anti-affine operations was $n = 3$, primarily due to redundancy in the attention weights. Thus, while increasing the number of anti-affine operations enhances long-range dependence, it may also introduce redundant information, thereby degrading the model's performance. Fig. 11 visualizes the different numbers of anti-affine operations.

C. Ablation of EA

1) Selection of β: The parameter β in (20) determines the importance DAEA-Net assigns to elevation information. We selected $\beta = 0, 0.25, 0.5, 0.75$, and 1 for experiments on the ISPRS and GML datasets. When $\beta = 0$, the gradient backpropagation learning does not occur through an additional loss function. The results, shown in Fig. 12, reveal that continuously adjusting the value of β on the ISPRS benchmark dataset fully integrates the elevation information, enhancing feature diversity and differentiation. On the simpler GML dataset, performance is lower with smaller β values at both ends and larger values in the middle. This suggests that a too-small β prevents the elevation module from adequately capturing vertical features, while a too-large β improves vertical discrimination but reduces horizontal discrimination. Therefore, a larger β can be used on datasets with complex structures, while a smaller β should be used for simpler scenarios. As seen in Fig. 12, β should be chosen based on the dataset's distribution.

2) Ablation Study of Core Components of EA: In Table IX, L represents the number of neighboring points selected to construct the weighted covariance matrix [see (9)], and $\omega\lambda$ indicates the weight coefficients for geometric structure discrimination. PP represents the updated point-pair information expression [see (14)], AE indicates the lossy reconstruction architecture of the encoding–decoding process, and RH_1 denotes the coded output features for enhancing absolute elevation information H_1 [see (15)]. From Table IX, the comparison of Models A, B, and our method shows that selecting six neighboring points to construct the covariance matrix is optimal. From Model C, using the three eigenvalues obtained from the weighted covariance matrix for geometric discrimination of neighboring points suppresses points with significantly different topologies, allowing for precise characterization of the elevation of the center point. Model F shows that introducing pairwise information enables the model to better capture elevation, improving its ability to distinguish different types of objects. Model E shows that without using the autoencoder for reconstruction loss learning, the model's OA and m$F1$ values decrease by 1.01% and 1.31%, respectively, indicating this method helps the model represent richer elevation information. Finally, Model D shows that incorporating absolute elevation information H_1 in the autoencoder's encoding phase through additive fusion reinforces absolute elevation, improving vertical discrimination across different categories.

Furthermore, the point-wise features from the penultimate fully connected layer were extracted, averaged to obtain a single-channel output, and converted into a grayscale image, as shown in Fig. 13. Fig. 13 clearly shows the contrast between roofs, trees, and the ground (the higher the object, the darker the color). This validates that our elevation module aids the network in expressing vertical directional features.

D. Other Experimentation

1) Ablation of Geometric Structure Building in LE: In Section III-A, the spatial representation $(p_i^k - p_i, \|p_i^k - p_i\|)$ and angular representation (θ_i^k, ϕ_i^k) were employed to construct the geometric structure, effectively capturing the translational and rotational invariance of the point cloud. This enhances the understanding of the point cloud's complex surface structure. To investigate how these representations affect the model, we conducted relevant experiments, and the results are presented in Table X.

The results show that adding either the polar angle θ_i^k or azimuthal angle ϕ_i^k individually provides additional geometric information to spatial representations and improves the network's characterization capabilities. Furthermore, incorporating both angular features enhances the characterization of

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TABLE VIII
EXPERIMENTS WITH DIFFERENT NUMBERS OF ANTI-AFFINE OPERATIONS

Numbers	Power	Low_veg	Imp_surf	Car	Fence	Roof	Facade	Shrub	Tree	OA	mF1
n=1	76.03	82.37	91.48	81.96	43.28	95.08	66.24	47.28	84.60	85.07	74.26
n=2	77.17	82.53	91.60	82.37	46.66	95.19	66.43	45.96	84.51	85.31	74.71
n=3	77.31	81.62	91.09	82.22	40.99	95.03	66.23	45.44	83.89	84.70	73.76

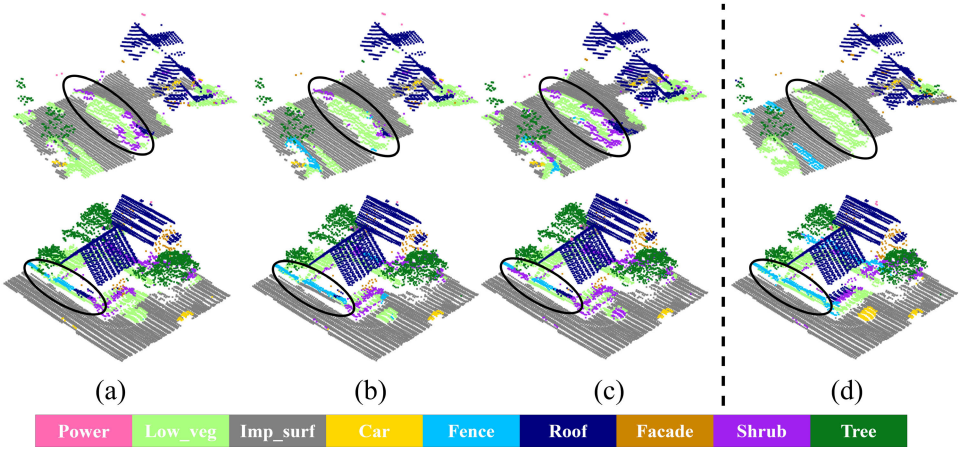


Fig. 11. Classification results of magnified regions with different number of repetitions on the ISPRS dataset. (a) Experiments with $n = 1$. (b) Experiments with $n = 2$. (c) Experiments with $n = 3$. (d) Ground truth.

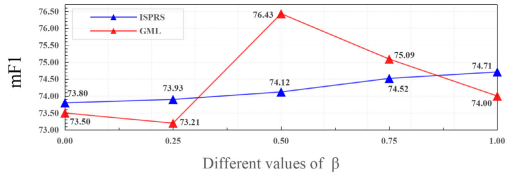


Fig. 12. Results for different β on the ISPRS dataset and GML dataset. The red line represents GML dataset results; the blue line, ISPRS dataset outcomes.

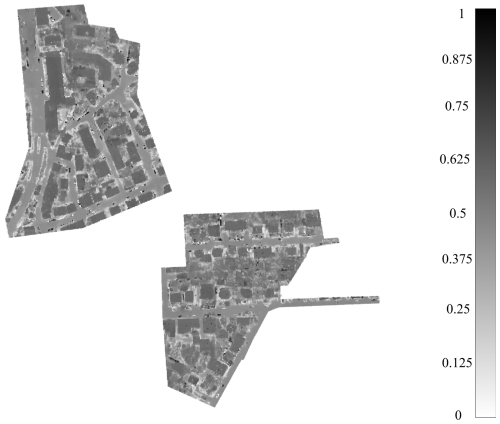


Fig. 13. Gray-scale maps for feature visualization in encoding–decoding architectures.

TABLE IX
ABLATION EXPERIMENTS FOR ELEVATION PERCEPTION

Model	L=5	L=6	L=7	$\omega\lambda$	PP	AE	RH_1	OA	mF1
Ours		✓		✓	✓	✓	✓	**85.31**	**74.71**
A	✓			✓	✓	✓	✓	85.15	74.54
B			✓	✓	✓	✓	✓	85.08	74.62
C		✓			✓	✓	✓	84.92	74.48
D		✓		✓	✓	✓		85.17	73.81
E		✓		✓	✓		✓	84.30	73.40
F		✓		✓		✓	✓	84.58	73.67

TABLE X
COMPARATIVE RESULTS OF GEOMETRIC STRUCTURE BUILDING METHODS IN LE

Model	Encoding Method	OA	mF1
A	$(p_i^k - p_i, \|p_i^k - p_i\|)$	84.50	74.00
B	$(p_i^k - p_i, \|p_i^k - p_i\|, \theta_i^k)$	84.91	74.46
C	$(p_i^k - p_i, \|p_i^k - p_i\|, \phi_i^k)$	84.68	74.81
D	$(p_i^k - p_i, \|p_i^k - p_i\|, \theta_i^k, \phi_i^k)$	85.31	74.71

TABLE XI
COMPARATIVE RESULTS UNDER DIFFERENT POOLING OPERATIONS IN LE

Model	Aggregation Operations	OA	mF1
A	sum-pooling	84.42	73.18
B	avg-pooling	84.09	73.51
C	max-pooling	85.31	74.71

geometric structures, thereby improving the model's performance.

2) Selection of Pooling Operations in LE: Table XI demonstrates the impact of different pooling operations for LE on model performance. According to Table XI, max pooling achieves the best performance, with OA values increasing by 1.22% and 0.89% and mF1 values increasing by 1.20% and 1.53%, respectively, compared to average pooling and summation pooling. The summation pooling operation computes the sum of all channels, which often homogenizes each centroid item within the convolutional aggregation framework. While average pooling provides more balanced performance and a better overall evaluation than summation pooling, it cannot

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TABLE XII
COMPARISON OF THE NUMBER OF MODEL PARAMETERS FOR DIFFERENT MODELS ON THE ISPRS BENCHMARK DATASET

Model	Params	OA	mF1
PointSIFT [17]	13.53	82.20	67.70
KPConv [35]	25.85	81.70	70.60
VD-LAB [49]	1.40	81.40	70.20
GraNet [40]	62.13	84.50	73.60
GAHD-Net [17]	14.80	84.50	73.20
DAEA-Net (Ours)	4.08	85.31	74.71

prevent the homogenization of each centroid item, leading to significantly lower performance compared to max pooling. Max pooling retains the most significant features in each channel and more accurately captures salient spatial variations in airborne LiDAR point cloud data.

3) Number of Model Parameters: In Table VI, we compared the parametric size between modules. Now, we compare it with other mainstream segmentation methods, and the specific results are shown in Table XII. From Table XII, we can see that our method has the best overall performance, with the highest OA and m$F1$ values and only 4 m network parameters.

VI. CONCLUSION

In this article, we introduce DAEA-Net, a novel network for semantic segmentation of airborne LiDAR point clouds, primarily composed of the CMAAA, EA, and UGA modules. The CMAAA module treats the center point as an object and strengthens long-range dependencies between neighbors by interactively fusing multiple features. To distinguish categories more effectively at the same height, the EA module introduces geometric structure discrimination, point-to-point information, and relative elevation. The UGA module adaptively fuses features at varying scales in the encoding–decoding layer, facilitating feature interaction and enhancing the model's expressive capacity. The proposed DAEA-Net enriches feature diversity and compensates for information loss in the sampling process, significantly improving the model's global characterization capability. To validate the proposed model, we tested it on the ISPRS and GML datasets, and the results indicate that compared to state-of-the-art methods, DAEA-Net demonstrates superior segmentation performance on OA and m$F1$ scores.

Nonetheless, our DAEA-Net still has certain limitations. For instance, the proposed local structure encoding only considers the relationship between a single neighboring point and the center point, neglecting the relationships among multiple neighboring points and the center point. This oversight leads to a loss of potential geometric spatial structure information among neighbors. The objective for the next stage is to reference traditional methods to develop an efficient approach to model the relationships among neighboring points, such as using the Gram matrix, thereby revealing more distinctive structural information.

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